

"Kestrel 1.5T: Catalyst efficiency below threshold from degraded upstream ox

Document: tsb-23-127

Area: bulletins

1 Condition

Some Kestrel 1.5T vehicles may illuminate the MIL with stored DTC P0420, catalyst system efficiency below threshold on Bank 1. The catalytic converter CAT-5500 may appear to underperform during the catalyst monitor, even though the converter itself is within specification. Customers usually report no drivability concern aside from the warning lamp. See Kestrel Emissions Service Manual :: Catalyst Efficiency Faults for the definition of P0420.

2 Cause

The root cause is a slow or degraded upstream oxygen sensor O2U-5100, whose lazy switching skews the air-fuel feedback and causes the catalyst monitor to misjudge converter efficiency. This produces a false-positive DTC P0420 without actual failure of catalytic converter CAT-5500. Confirm sensor response and compare upstream versus downstream signals per Kestrel Emissions Service Manual :: Upstream and Downstream Oxygen Sensors before condemning the converter described in Kestrel Emissions Service Manual :: Catalytic Converter (Bank 1 and Bank 2) .

3 Repair Procedure

Replace the upstream oxygen sensor O2U-5100 following procedure PROC-O2-SENSOR as detailed in Kestrel Emissions Service Manual :: Oxygen Sensor Replacement Procedure . Review Kestrel Emissions Service Manual :: Oxygen Sensing for connector routing and torque before installation. After replacement, clear DTC P0420 and run the catalyst and oxygen sensor readiness monitors to verify the repair, following Kestrel Emissions Service Manual :: Emissions System Diagnosis . Do not replace the converter unless the monitor still fails with a verified-good sensor installed.